Replication of the social information transmission findings by Mesoudi, Whiten & Dunbar (2006, British Journal of Psychology)

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Introduction

Mesoudi, Whiten, and Dunbar (2006) asked whether some kinds of information survive transmission better than others. They used the transmission chain method, in which one participant's recall becomes the subsequent participant's stimulus, so that information fidelity and loss can be tracked as it travels. In Experiment 2, the first participant in a four-person chain read four short paragraphs, wrote down what they remembered, and then that written recall was given

to the next participant in the chain. The four paragraphs matched in length and structure, each containing 14 propositions across 5 sentences and written as a chain of events linked causally and temporally so that no paragraph was more coherent than another. However, the content type of these paragraphs differed: gossip about third-party social relationships, ordinary non-gossip social interactions, individual behavior, and physical events. Paragraph order was counterbalanced. Every generation of the chain's recall was scored against the original paragraph, using Kintsch (1974) propositional analysis. The target finding for this replication is the effect of content type on how much survived, $F(3, 24) = 15.29$ for the total number of propositions recalled. Gossip and non-gossip social information survived transmission better than individual behavior or physical events, and the contrast between gossip and non-gossip social information was null at $F(1, 9) = 0.16$, which the authors interpreted as being due to the presence of people interacting with each other rather than the scandalous nature of the gossip content.

My current research asks how descriptions of people change as they pass between people, analyzing the compression and content convergence of social information across generations of transmission. I chose to replicate Mesoudi et al.'s experiment because it is more or less a rudimentary version of the study I am about to run. Their version includes chains of four participants, written recall from one participant to the next, each chain treated as an independent unit of analysis. However, while Mesoudi et al. are interested in how differing content in general survives transmission, I am interested in how differing *social* content (relationships, traits, actions, locations) about a single person survives transmission when presented all at once in one recall. Their four paragraphs were designed to vary widely from non-social to social in content, so what this replication will validate is a multi-generational chain as a sensitive enough instrument to detect content effects at all. If it is unable to separate narratives that differ this greatly, it will likely also not be able to separate smaller differences in social content such as relationships and traits.

I plan to replicate Experiment 2 by running online participants from UCSD's undergraduate participant pool (SONA). The experimental task will be run through a web application I am building for my own chain study, which presents a previous generation's text and collects the next generation's written recall. Each participant will read the four paragraphs presented in the original study once in a counterbalanced order. Then, they will write down what they remember, not constrained by time limit and without being presented with a distractor task. Participants will not be told that this is a recall test, that the text came from another participant prior to them, or that their recall will be passed on. I will manually mediate transmission, ensuring an equal number of participants per chain and asynchronous participation.

There will be four primary challenges. First, the original study ran in person on printed paper, where rereading the page was prevented by physically turning the page. My web application will hide the text before the recall box appears to combat this. Online participants may also be less attentive or use outside sources to either offload information or generate responses. To address this I intend to include a minimum reading time, attention checks, and screening for responses that appear to have been assisted by AI. Second, the original scored recalls by

hand, which is not feasible to reproduce at scale. Instead, I plan to score each recall against the original paragraphs directly using BERTScore and an LLM at temperature 0, both of which Martinez (2024) found to reach an r of at least .97 against manual scoring of story recall. I also will hand code a subset using Kintsch (1974) propositional analysis, so that a failure to reproduce the effect is distinguishable from a failure due to the scoring. Third, the four paragraphs are matched on propositions but not on words (which can range from 54-66), and they differ in how much their wording repeats, so a content effect could potentially come from the metric rather than from transmission. To this end I will score the four original paragraphs against each other before collecting any data to see what happens to the metric when nothing has been transmitted yet. Finally, the effect I am focusing on is smaller than Experiment 1's, the within-social contrast is null, and no power analysis is reported in the original paper, so I will determine the number of chains from a simulation rather than copying the original's 10.

From this project, I am hoping to reach four learning goals. First, I have never used SONA to recruit participants. I am only familiar with Prolific, so this is a good way to test the waters with SONA and validate it as a reliable subject pool for future research, especially as Prolific expenses are increasing. Second, I will be able to validate automated recall scoring against hand coded data. So far, I have built a tool that extracts person mentions from unnamed relational data in my own paradigm, but I have not yet validated nor expanded this tool in terms of scoring whole recalls. Third, I would like to explore power simulation for designs where the chain rather than the participant is the independent unit. This is the same question as how many chains my own study will need. Fourth, more broadly, I aim to gain more familiarity with reading and running repeated-measures designs of this kind.

I would wager this project as an entirely feasible build for the quarter. The stimuli are published, the experimental task is asynchronous and mediated by myself, so I never need concurrent participants. I already have a good starting point on a web app that presents a generation's text and collects the next generation. Additionally, the original study had 10 chains of four participants, which is entirely doable in a week or two and can be reasonably expanded, pending the results of the power analysis.

Response to feedback. I have changed projects since the original proposal, from Panela et al. (2025) on LLM event segmentation to Mesoudi, Whiten, and Dunbar (2006). The original proposal was built around validating an automated chunking method, but following a meeting with my advisor, that research goal was deferred. I instead would find it most pertinent and beneficial to run a diffusion chain paradigm, which is what the study I am about to run uses. Four of the five feedback points from the earlier proposal carry over, so I will address them here in light of my new proposal instead of dropping them.

Regarding examining how the measure works, the carryover to my current replication is that an automated similarity score may behave differently across the four paragraphs, completely unrelated to transmission (outlined as the third challenge in the challenges section above). There is no repository for this paper published in 2006, so I will email Mesoudi to ask for their original coding sheets.

Regarding keeping at least one original component so that failure of methods can be distinguished from failure of effect, I am hand coding a subset of recalls using Kintsch (1974) propositional analysis rather than entirely replacing the original with automated scoring.

Regarding planning exclusion criteria up front, since online participants behave differently than in person ones, I am including minimum reading time, attention checks, and screening for AI assisted responses (outlined as the first challenge in the challenges section above).

Regarding adding an exploratory section tied to my own research, specifically adding a fifth narrative that is drawn from a spoken recall, I would rather keep that separate. Adding a fifth narrative to the experiment changes how much there is to remember, so the design would no longer be a replication of Experiment 2. Additionally, a written recall of a spoken excerpt scored against its source is not really relevant to any comparisons my own study makes, since my study exclusively uses spoken recall and measures convergence between generations in addition to fidelity to the original. I will run that comparison there instead, where it fits the design more neatly.

> **i** Links
>
> Repository: https://github.com/kyaneos/mesoudi2006
> Original paper: 2006_mesoudi_social_information_bias.pdf (Retrieved on Oct 12, 2026 from doi.org/10.1348/000712605X85871)

Methods

Power Analysis

Original effect size, power analysis for samples to achieve 80%, 90%, 95% power to detect that effect size. Considerations of feasibility for selecting planned sample size.

Planned Sample

Planned sample size and/or termination rule, sampling frame, known demographics if any, preselection rules if any.

Materials

All materials - can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Procedure

Can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Analysis Plan

Can also quote directly, though it is less often spelled out effectively for an analysis strategy section. The key is to report an analysis strategy that is as close to the original - data cleaning rules, data exclusion rules, covariates, etc. - as possible.

Clarify key analysis of interest here You can also pre-specify additional analyses you plan to do.

Differences from Original Study

Explicitly describe known differences in sample, setting, procedure, and analysis plan from original study. The goal, of course, is to minimize those differences, but differences will inevitably occur. Also, note whether such differences are anticipated to make a difference based on claims in the original article or subsequent published research on the conditions for obtaining the effect.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or "none".

Results

Data preparation

Data preparation following the analysis plan.

Confirmatory analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.